EXPERIMENT 5

Using Raptor – Draw and validate a flowchart for a given string of lower case English alphabets. One can choose any two characters in the string and replace all the occurrences of the first character with the second character and replace all the occurrences of the second character with the first character. Using Raptor draw and validate the flowchart lexicographically smallest string that can be obtained by doing this operation at most once.

Raptor CO1

Objective (Divyadarshini ,danustpriya)

To design and validate a RAPTOR flowchart that transforms a given lowercase string into the

lexicographically smallest possible string by swapping two characters at most once.

Algorithm

1. Read string str
2. Find the first position i where a smaller character exists later
3. Select two characters c1 and c2
4. Replace all c1 with c2 and all c2 with c1
5. Display the modified string

Step-by-Step RAPTOR Instructions

□□ Step 1: Start RAPTOR

Open RAPTOR → Click New Flowchart

□□ Step 2: Input String

Insert Input symbol

Input str

□□ Step 3: Initialize Variables

Insert Assignment symbol

n ← length(str)

found ← false

□□ Step 4: Outer Loop (i)

Insert Loop symbol (For loop)

fori ← 1 to n

□□ Step 5: Inner Loop (j)

Inside loop:

for j ← i+1 to n

□□ Step 6: Decision to Find Smaller Character

Insert Decision symbol

ifstr[j] <str[i] and found = false

□□ Step 7: Store Swap Characters

Inside IF:

c1 ← str[i]

c2 ← str[j]

found ← true

□□ Step 8: End Both Loops

Exit loops once found = true

□□ Step 9: Replace Characters in String

Insert loop:

for k ← 1 to n

Inside loop:

ifstr[k] = c1

str[k] ← c2

else if str[k] = c2

str[k] ← c1

end if

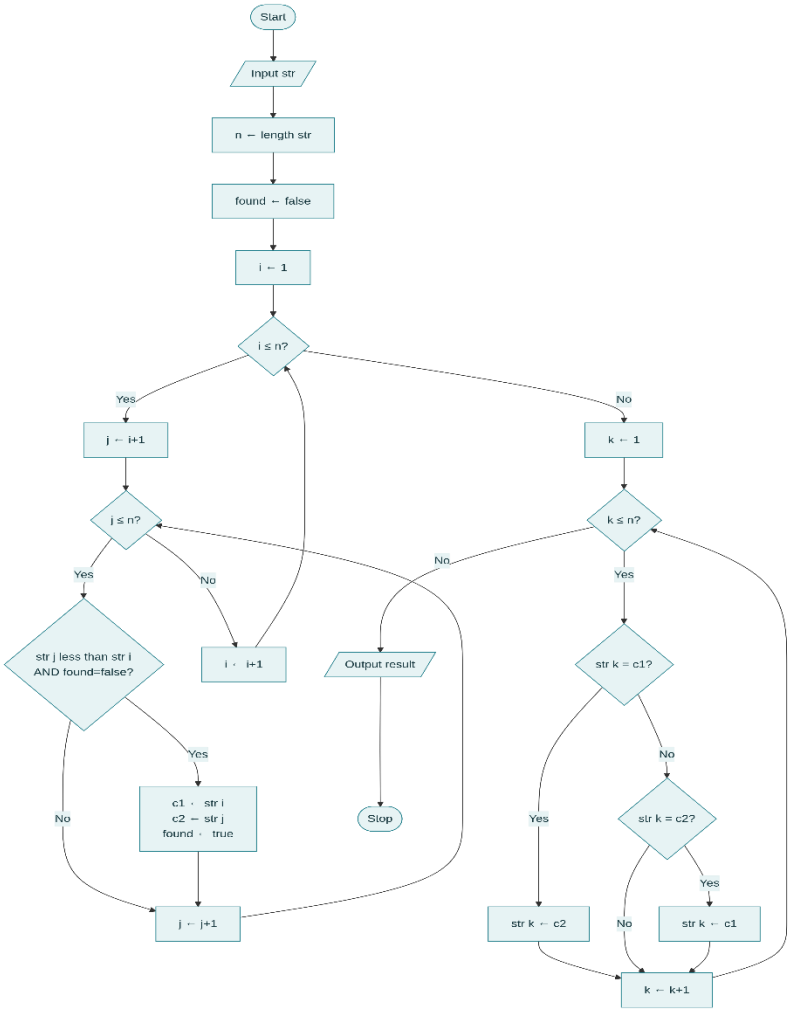
□□ Step 10: Output Result

Insert Output symbol

Output "Lexicographically smallest string = ", str

□□ Step 11: Stop

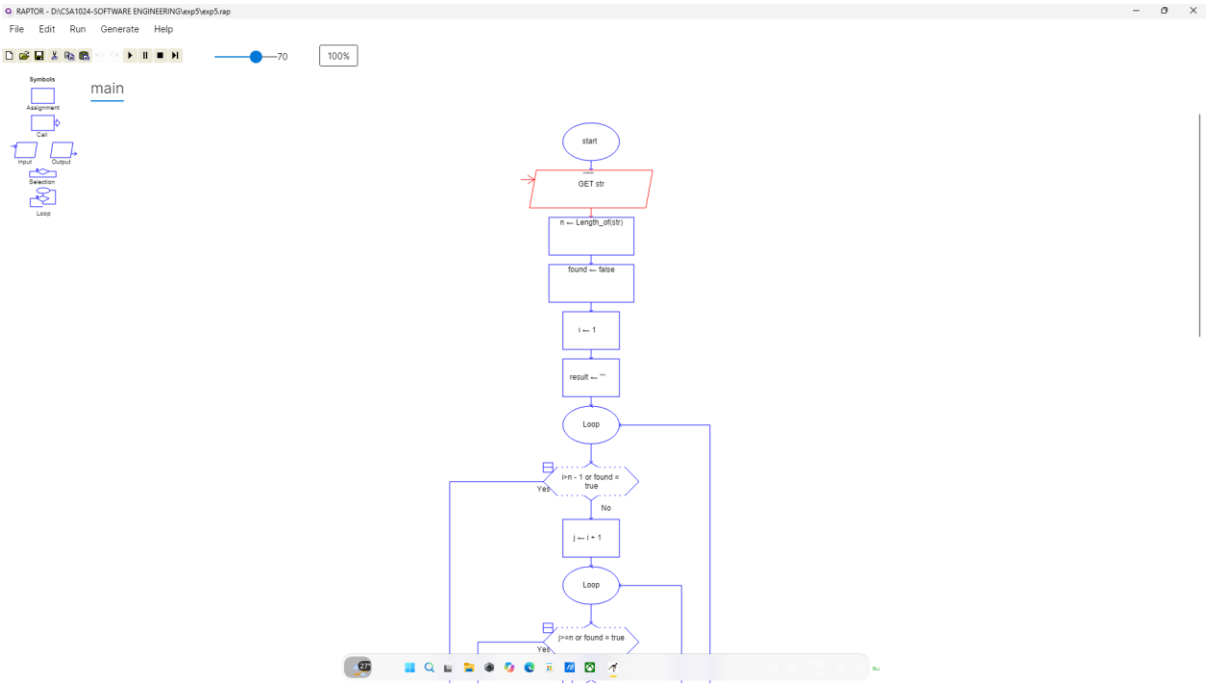
Flowchart

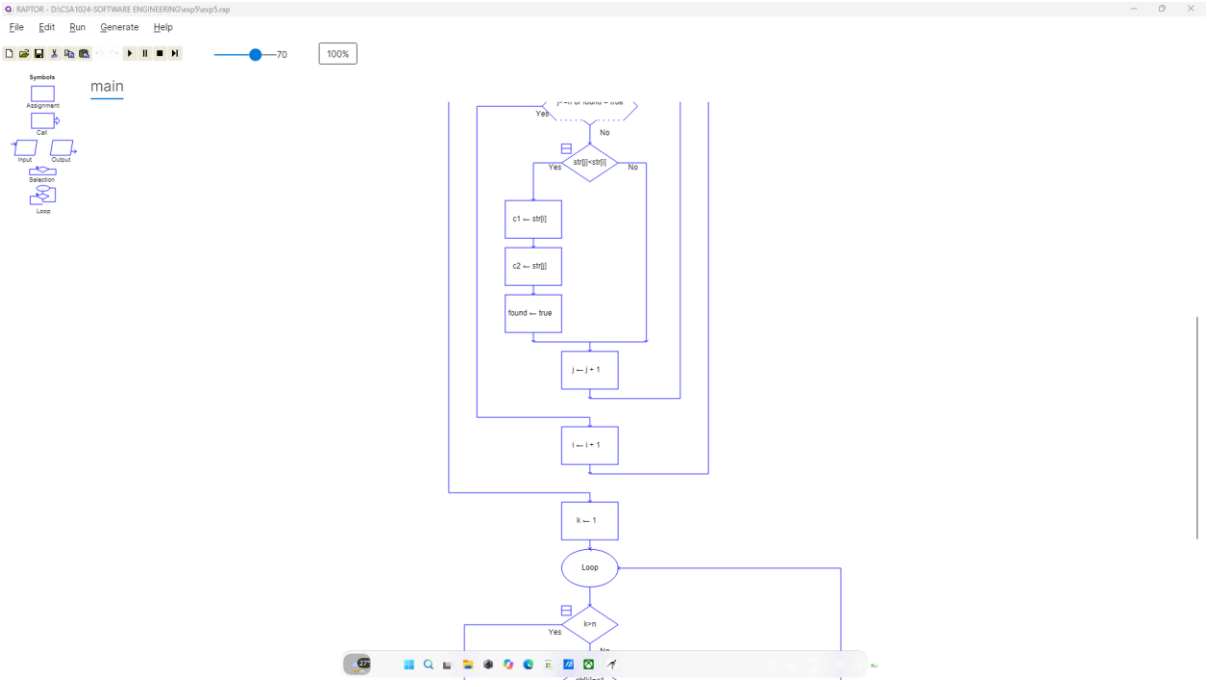


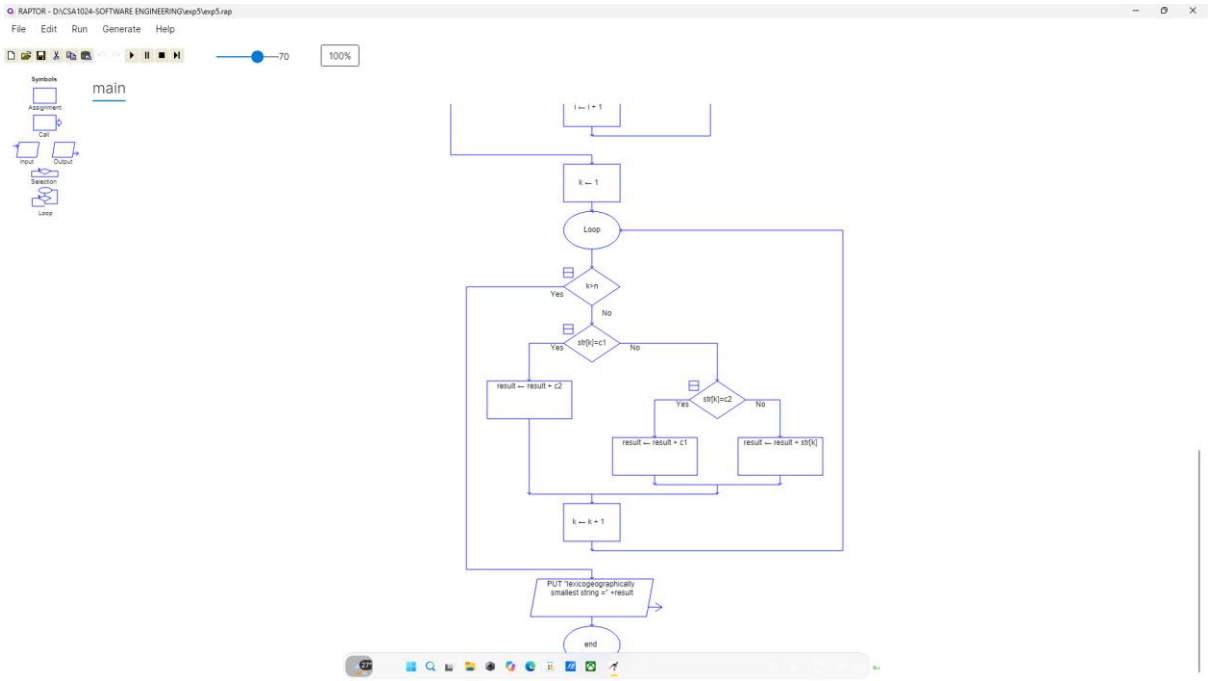
Result:

Thus, a RAPTOR flowchart was successfully designed and validated to generate the

lexicographically smallest string using at most one character swap.


RAPTOR - D:\CSA1024-SOFTWARE ENGINEERING\exp5\exp5.rap
File Edit Run Generate Help
100%
Symbols
Assignment
Call
Input
Output
Selection
Loop
main
start
GET str
n ← Length_of(str)
found ← false
i ← 1
result ← ""
Loop
i>n - 1 or found = true
Yes
No
j ← i + 1
Loop
i>=n or found = true
Yes


RAPTOR - D:\CSA1024-SOFTWARE ENGINEERING\exp5\exp5.rap
File Edit Run Generate Help
100%
Symbols
Assignment
Call
Input
Output
Selection
Loop
main
Yes
No
str[j]<str[i]
Yes
No
c1 ← str[i]
c2 ← str[j]
found ← true
j ← j + 1
i ← i + 1
k ← 1
Loop
k>n
Yes


RAPTOR - D:\CSA1024-SOFTWARE ENGINEERING\exp5\exp5.rap
File Edit Run Generate Help
70
100%
Symbols
Assignment
Call
Input
Output
Selection
Loop
main
i ← i + 1
k ← 1
Loop
k>n
Yes
No
str[k]=c1
Yes
No
result ← result + c2
str[k]=c2
Yes
No
result ← result + c1
result ← result + str[k]
k ← k + 1
PUT "lexicogeographically smallest string =" +result
end